

Computational Wind Engineering

Introduction

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The aim of the course is to provide you with an understanding of the fundamental principles of Computational Wind Engineering

Computational/Wind Engineering

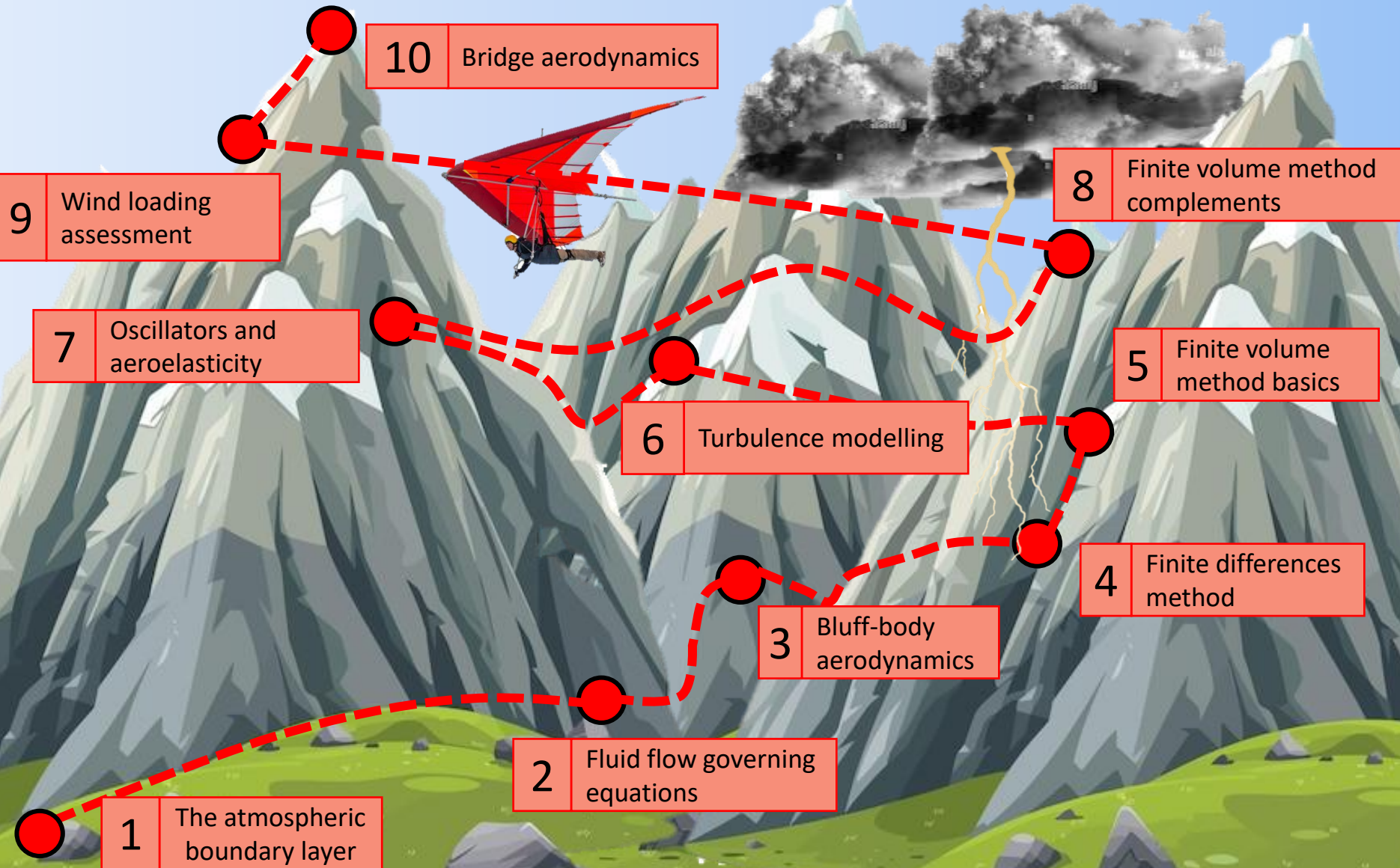
Our efforts will focus on deriving the main concepts and guiding principles of Computational Wind Engineering (CWE) from **fundamental principles**.

Map of the course

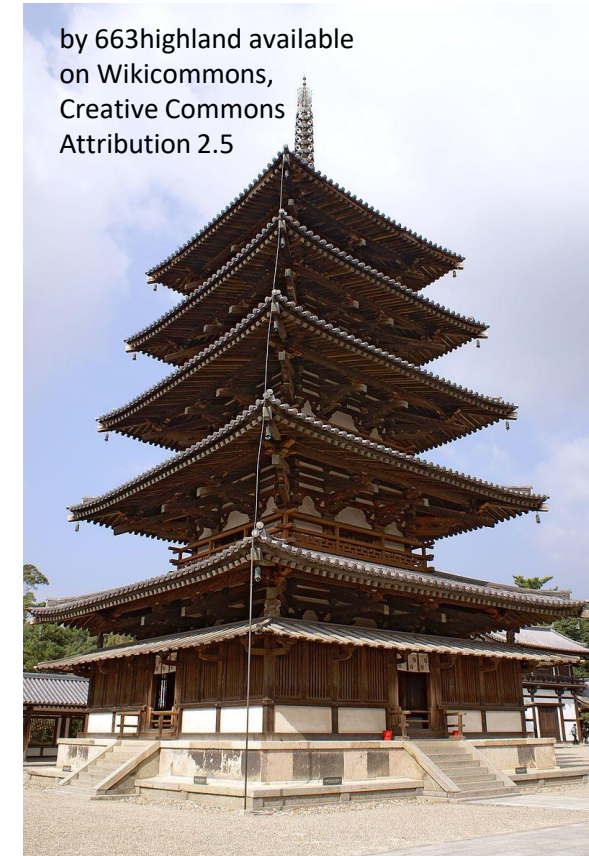
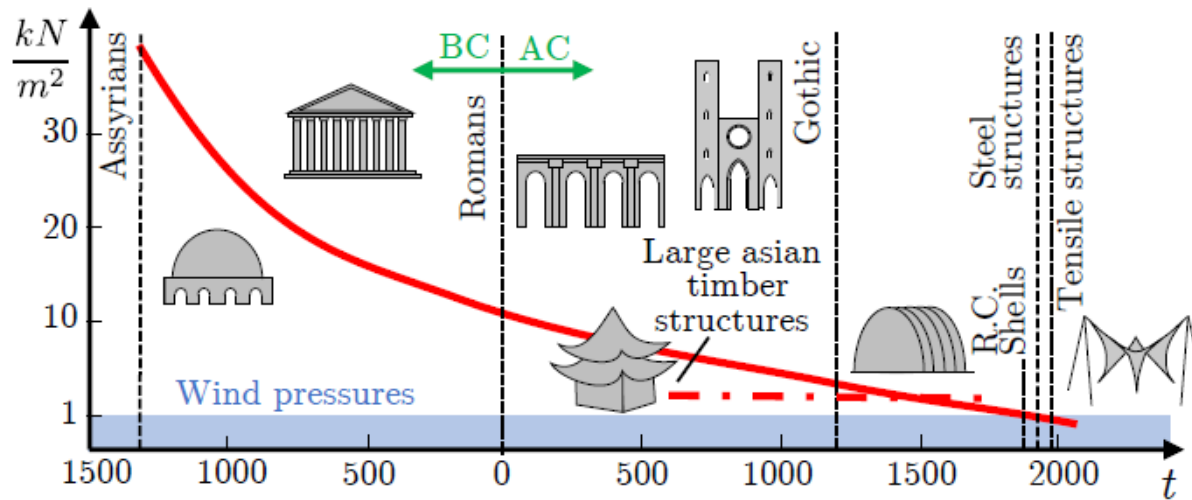
Wind Engineering

Fluid Dynamics

Numerical methods



The importance of wind is closely related to the structural typology.



Light structures are characterised by the fact that accidental loads are comparable of dominate dead loads.

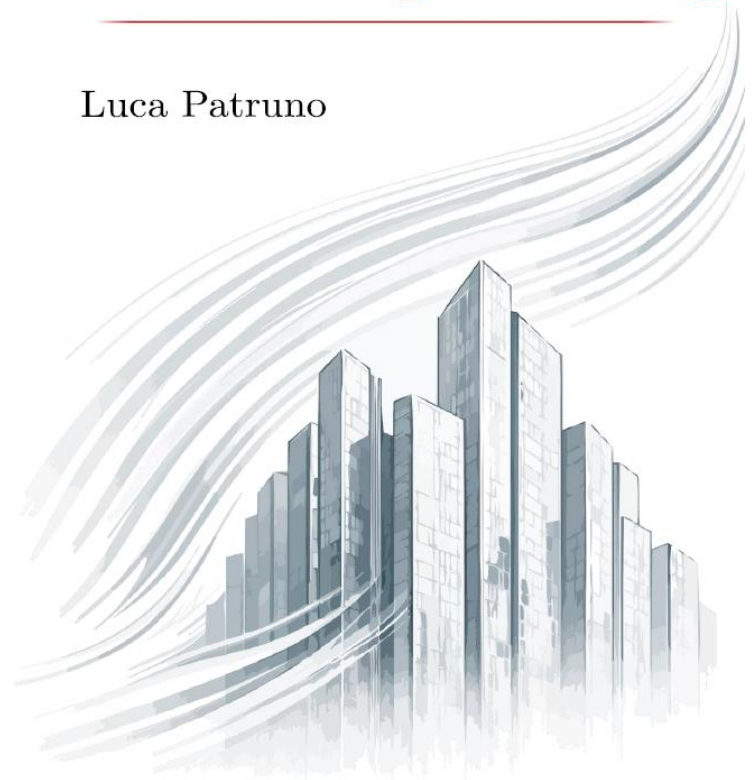
These presentations a concise summary of the material from which a book, now available on Amazon, was derived.

The presentations do not cover all the content of the book and **often skips explanations and details.**

The material is provided in its original form, as used for teaching at the University of Bologna.

Essential Concepts in
**Computational
Wind Engineering**

Luca Patruno



As anticipated, these slides should not be considered as a satisfactory summary of a university course (they are not projected during the course), nor as a summary of the companion book.

They have been created mainly to fulfill two objectives:

- 1) provide the students with a set of notes to be completed during lectures (avoiding them to take notes continuously);
- 2) provide the instructor with a detailed track around which lectures can be organised.

The final chapter of the book, focused on applications, is omitted as it requires hands on exercitations.